



Reg. No:

--	--	--	--	--	--	--	--	--	--	--

SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' GRADE Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2024

B.E - COMPUTER SCIENCE AND ENGINEERING

20CS259 - FOUNDATIONS OF BLOCK CHAIN TECHNOLOGY

Answer ALL questions

Duration: 2 Hrs 30 min

Maximum: 70 Marks

PART - A

(5x 1 = 5 Marks)

1. **CO1** Blockchain is a peer-to-peer _____ distributed ledger technology that makes the records of any digital asset transparent and unchangeable.
a) Decentralized b) Demanding c) Secure d) Popular
2. **CO1** What is a Nonce in a Blockchain network?
a) A random value used to vary the value of the hash
b) A fixed value used to verify the authenticity of transactions
c) A fixed value used to modify the records
d) A fixed value used to delete the records
3. **CO2** In block chain, which tree stores all the transactions in a block by producing digital fingerprint of the entire set of transactions?
a) AVL b) Merkle c) Binary d) Red Black
4. **CO3** What consensus mechanism does Ethereum plan to transition to in order to improve scalability and energy efficiency?
a) Proof-of-Work (PoW) b) Proof-of-Authority (PoA)
c) Proof-of-Stake (PoS) d) Delegated Proof-of-Stake (DPoS)
5. **CO3** Corda is a blockchain platform specifically designed for which sector?
a) Healthcare b) Banking and finance
c) Supply chain management d) Social media

PART - B

(5x 3 = 15 Marks)

6. **CO1** What are the different types of Blockchain?
7. **CO1** Differentiate Proof-of-work v/s Proof-of-stake protocol
8. **CO2** List out the advantage of Cryptocurrency.
9. **CO3** Define Ethereum (ETH).
10. **CO3** State the primary advantage of using SNARKs in blockchain systems.

PART – C

(5 x 10 = 50 Marks)

Q.No: 11 Compulsory Question

Q.No: 12 to 17 Answer any FOUR Questions

Compulsory Question:

11. **CO1** Explain specific challenges and limitations of achieving consensus in blockchain networks compared to traditional distributed systems. (10)
12. **CO1** i) How does Byzantine adaptation to internal failure (BFT) apply to decentralized networks? (5)
CO1 ii) Explain how fairness and liveness can be achieved in blockchain technology. (5)
13. **CO1** i) Explain RLA model in blockchain technology. (5)
CO1 ii) Explain the concept of Proof of Work (PoW) as a Random Oracle in blockchain technology and its significance in achieving decentralized consensus. (5)
14. **CO2** Design a blockchain-based voting system for a local election, incorporating Merkle trees to ensure the integrity and security of the voting process. Evaluate how Merkle trees can be employed to efficiently verify the authenticity of votes, deter tampering with voting records, and offer transparent and auditable results to stakeholders. Propose solutions to potential challenges or considerations in implementing Merkle trees in the voting system. (10)
15. **CO2** Define a double-spend attack within the context of blockchain technology, and explain its significance as a threat to the integrity of transactions in decentralized networks. (10)

16. **CO3** i) Explain smart contract, and how does it function in blockchain technology (5)
- CO3** ii) Analyze the trade-offs and challenges associated with implementing zero-knowledge proofs in blockchain networks, including computational overhead, verification complexity, and adoption barriers. (5)
17. **CO3** Discuss how does the use of a blockchain-based system help in reducing the cost and complexity of the land registry and smart toll systems? (10)
